

A database for managing animal shelters

Both parts of the exam preparation use this scenario. A database named *PetShelters* stores information about shelters, which house pets. These pets can be adopted by people, who have to reserve them first. Below you can find the relation schema for the four relations in the database, as well as the concrete relation instance as it is stored in the database. Additionally an explanation in natural language is given below that.

Shelter(shelterId, city)

shelterId	city
0	Lyngby
1	Sorgenfri
2	Virum

Person(personName, city)

personName	city
Dorit	Sorgenfri
Eli	Gentofte
Tim	Lyngby
Vanessa	Sorgenfri

Pet(petName, age, breed, shelterId)

petName	age	breed	shelterId
Dorit	4	Parrot	2
Einstein	5	Pug	0
Gonzales	1	Mouse	1
Gordon Moore	10	British Shorthair	0
Harald Keramiker	9	British Shorthair	2
Jonathan	193	Seychelles Giant Tortoise	1
Julius Caesar	11	British Shorthair	0
Kaa	21	Indian Python	2
Mr. Fluffy	2	Poodle	0
Remy	6	Rat	2
Tim	3	Golden Retriever	2
Usain Bolt	4	Pug	2

reservedFor(personName, petName, reservedDate)

personName	petName	reservedDate
Dorit	Gordon Moore	2026-01-13 00:00:00
Dorit	Harald Keramiker	2026-01-13 00:00:00
Dorit	Julius Caesar	2026-01-13 00:00:00
Eli	Usain Bolt	2026-03-27 00:00:00
Tim	Mr. Fluffy	2026-03-17 00:00:00
Tim	Usain Bolt	2026-03-14 00:00:00
Vanessa	Kaa	2026-02-14 00:00:00
Vanessa	Remy	2026-02-14 00:00:00

Shelter: Each shelter has a unique *shelterId* and the *city* in which the shelter is located.

Person: Every person has a unique *personName* and the *city* where they live.

Pet: Each pet is an animal with a unique *petName*, its *age* (integer), the *breed* of the animal, and the *shelterId* of the shelter in which it currently is housed.

reservedFor: A pet can be reserved for adoption by a person, in which case it is recorded in this relation. Each reservation (row in *reservedFor*) has a unique combination of *personName* and *petName* and additionally records the date of the reservation (*reservedDate*).

Question 1

Write an SQL query that returns a table containing the breed and average age of pets in the Lyngby shelter.

Question 2

Write an SQL query that returns a table containing the names, which are shared between Pets and Persons.

For example, if there is both a pet named Julian and a person named Julian, the table would contain this name.

Question 3

Write an SQL query that returns a table containing the name and total number of reservations a pet has.

Name the total number of reservations *numReservations* and order the resulting table by it, in descending fashion.

Important note: Make sure that every pet appears in the table, even those that might not have a reservation!

Question 4

Create a trigger that raises a signal, if a reservation is insert into table *reservedFor*, where the reservation date is in the future. Make sure to use a fitting MySQL error number, sql state, and message that describes the problem to the user.

Question 5

Write an example SQL query that is prevented by your trigger.